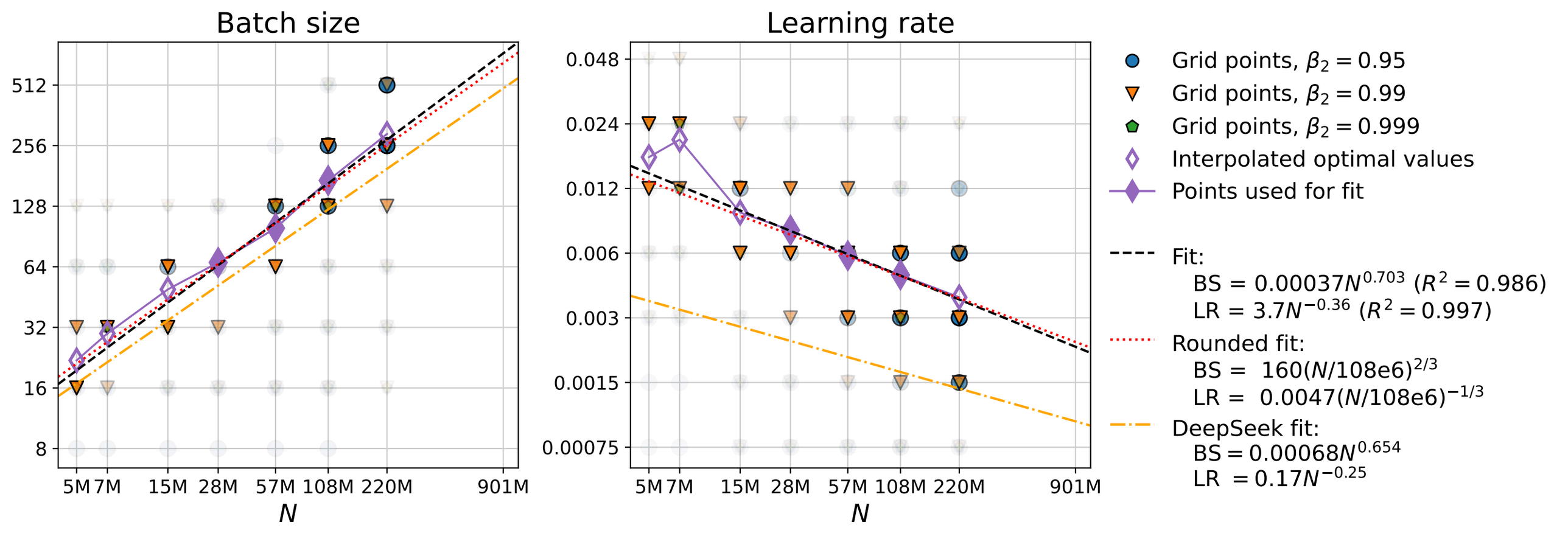
Batch size
512
256
128
64
32
16
8
5M 7M
15M
28M
57M
108M
220M
901M
N
Learning rate
0.048
0.024
0.012
0.006
0.003
0.0015
0.00075
5M 7M
15M
28M
57M
108M
220M
901M
N
Grid points, $\beta_2 = 0.95$
Grid points, $\beta_2 = 0.99$
Grid points, $\beta_2 = 0.999$
Interpolated optimal values
Points used for fit
Fit:
BS = $0.00037N^{0.703}$ ($R^2 = 0.986$)
LR = $3.7N^{-0.36}$ ($R^2 = 0.997$)
Rounded fit:
BS = $160(N/108e6)^{2/3}$
LR = $0.0047(N/108e6)^{-1/3}$
DeepSeek fit:
BS = $0.00068N^{0.654}$
LR = $0.17N^{-0.25}$